Multivariate Time-Series Prediction Using Deep Learning

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1. Introduction

1.1 Problem Overview

Predicting household appliance energy consumption is challenging because electricity usage changes throughout the day. This project uses the UCI Appliances Energy Prediction dataset, containing 19,735 observations recorded every 10 minutes from a house in Belgium. The dataset includes appliance energy consumption, indoor temperature, humidity, and outdoor weather measurements.

1.2 Project Objectives

The objective is to develop and optimise deep learning models for short-term appliance energy forecasting, using data preprocessing, feature engineering, and model evaluation. Traditional machine learning models are also used as baselines to compare performance and identify the most effective approach.

Data Insights

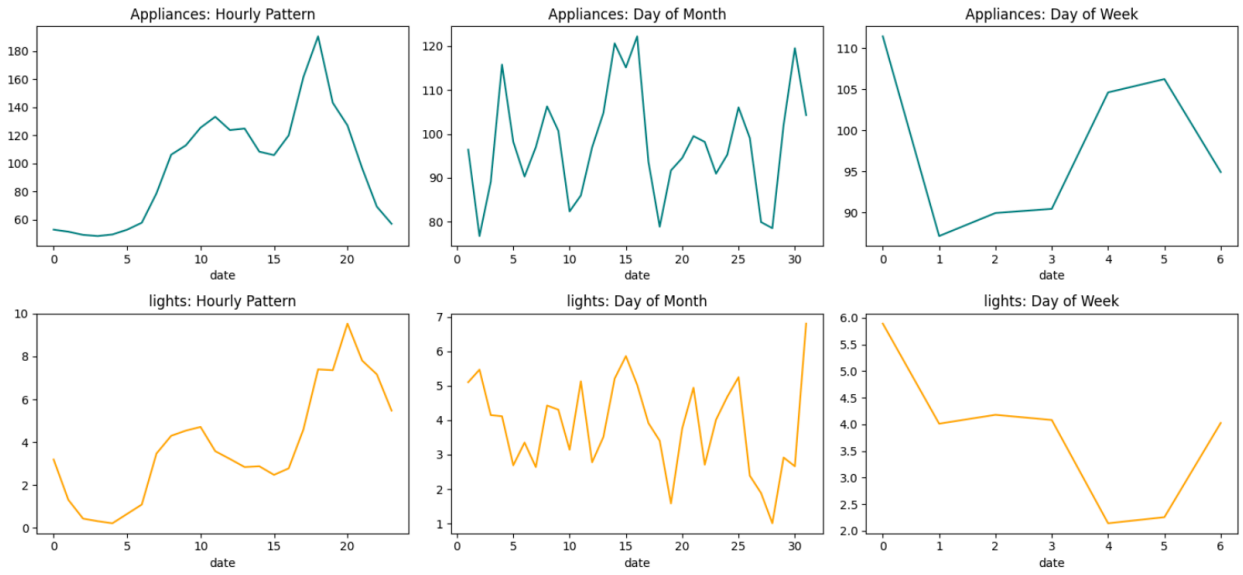
2.1 Dataset Overview

This project uses the UCI Appliances Energy Prediction dataset (Candanedo et al., 2017), collected from a low-energy house in Belgium. It contains 19,735 observations recorded every 10 minutes over 137 days, from 11 January to 27 May 2016.

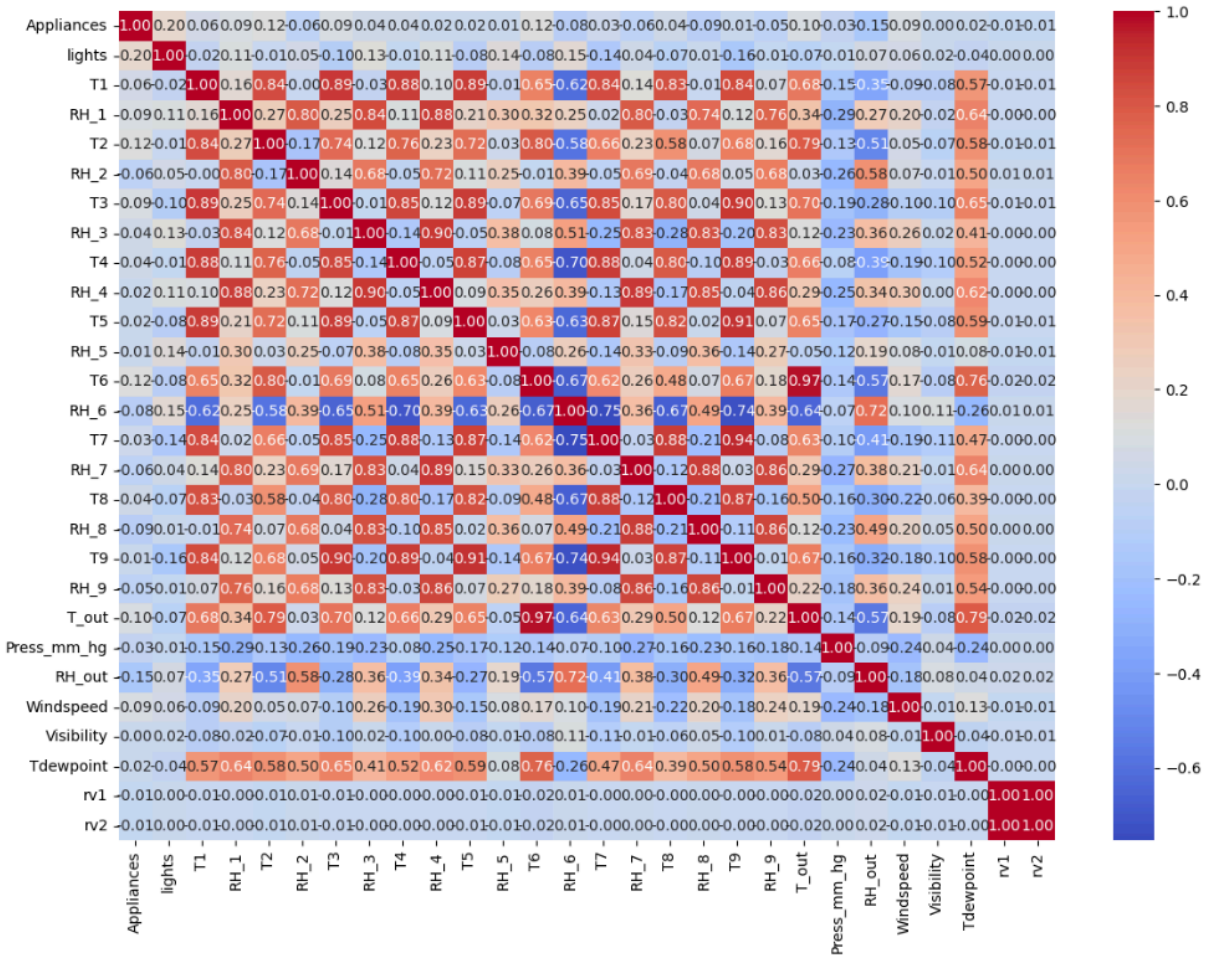
The target variable is appliance energy consumption (Appliances), measured in Wh. The dataset also includes lighting consumption, temperature and humidity readings from nine indoor sensor locations, outdoor weather measurements, and two random variables (rv1 and rv2) included for feature selection experiments.

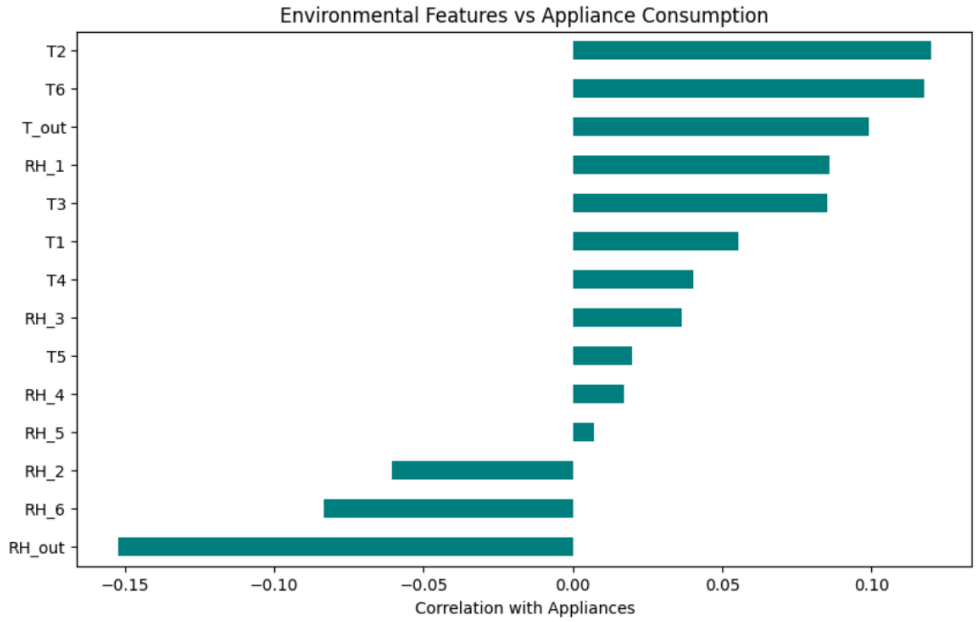
Rows - 19,735
Columns - 29 (original dataset)
Date range - 2016-01-11 17:00 to 2016-05-27 18:00
Duration - 137 days
Sampling Interval - 10 minutes
Duplicate rows - 0
Missing values - 0

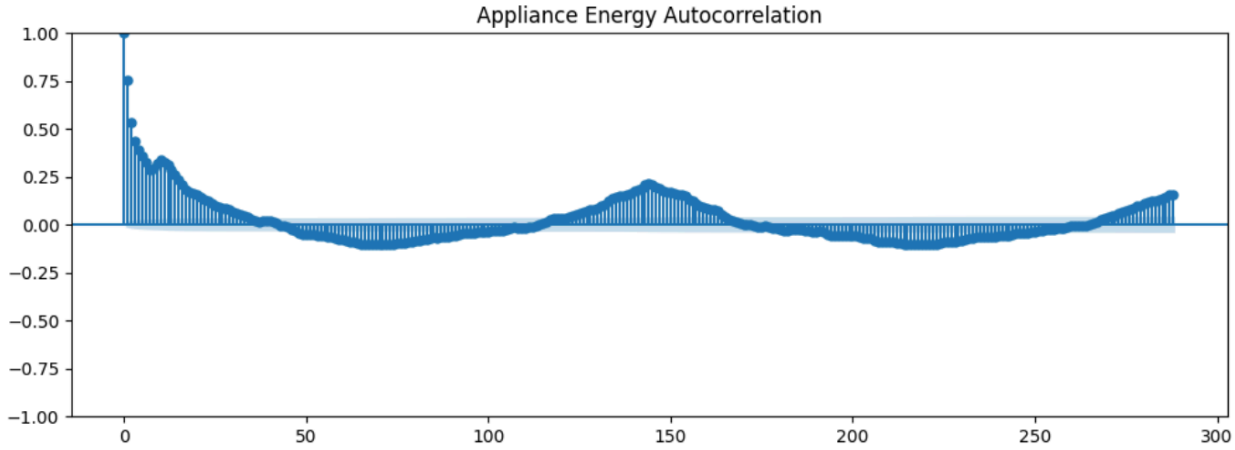
2.2 Exploratory Data Analysis

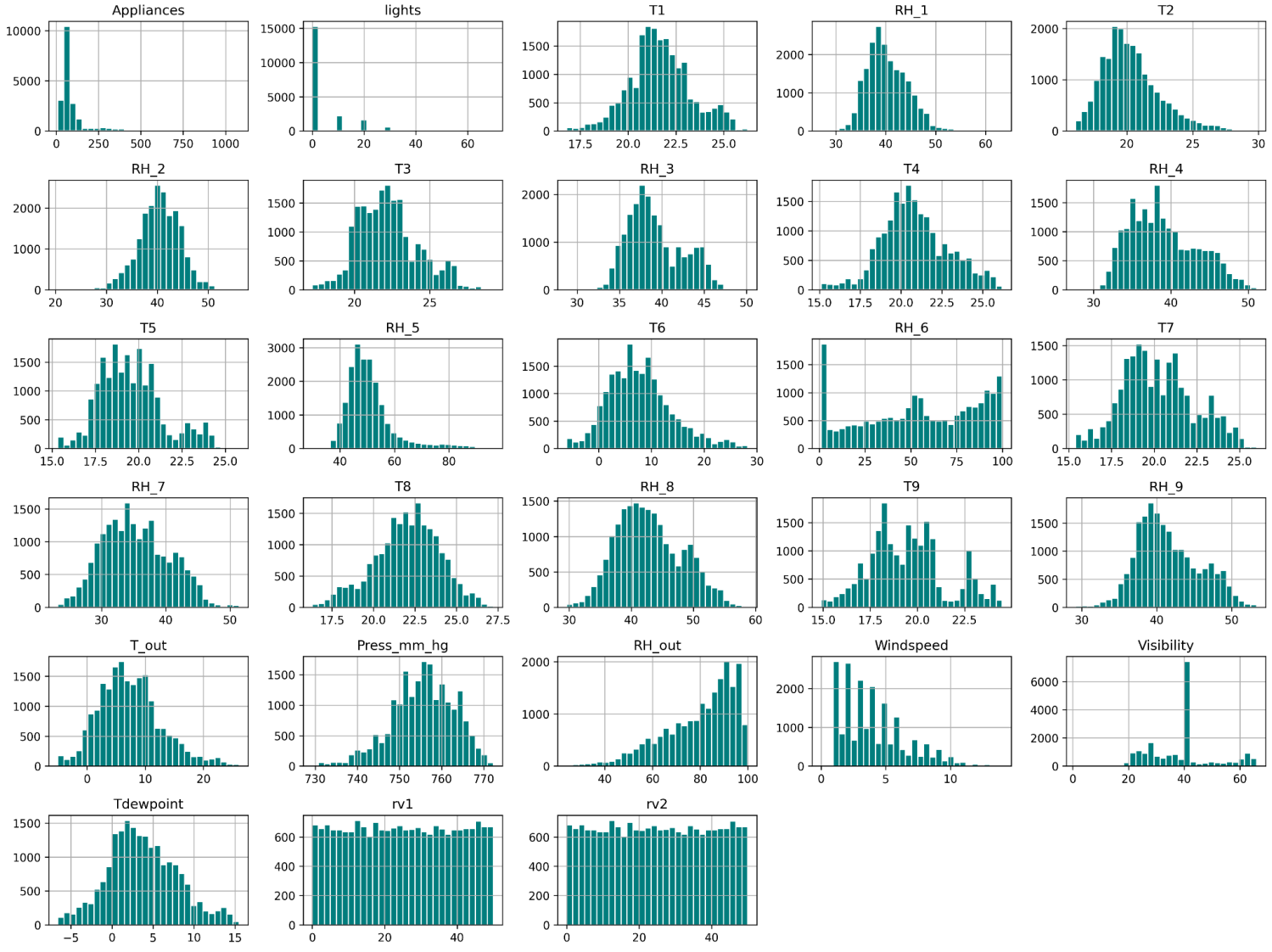


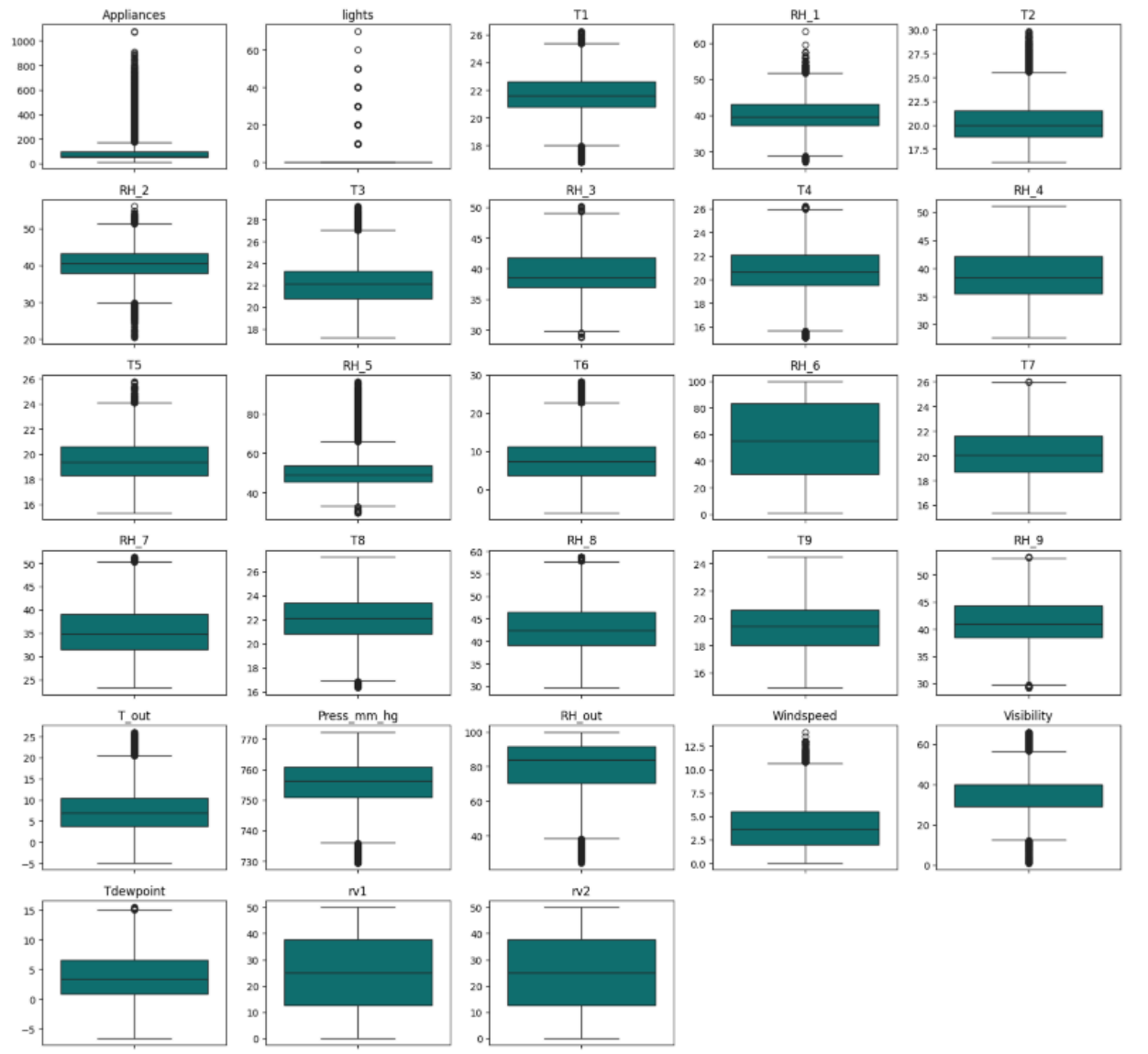
Key finding -


Appliances
lights
T1
RH_1
T2
RH_2
T3
RH_3
T4
RH_4
T5
RH_5
T6
RH_6
T7
RH_7
T8
RH_8
T9
RH_9
T_out
Press_mm_hg
RH_out
Windspeed
Visibility
Tdewpoint
rv1
rv2
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6


Environmental Features vs Appliance Consumption
T2
T6
T_out
RH_1
T3
T1
T4
RH_3
T5
RH_4
RH_5
RH_2
RH_6
RH_out
−0.15
−0.10
−0.05
0.00
0.05
0.10
Correlation with Appliances


Appliance Energy Autocorrelation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
50
100
150
200
250
300


Appliances
lights
T1
RH_1
T2
RH_2
T3
RH_3
T4
RH_4
T5
RH_5
T6
RH_6
T7
RH_7
T8
RH_8
T9
RH_9
T_out
Press_mm_hg
RH_out
Windspeed
Visibility
Tdewpoint
rv1
rv2


Appliances
lights
T1
RH_1
T2
RH_2
T3
RH_3
T4
RH_4
T5
RH_5
T6
RH_6
T7
RH_7
T8
RH_8
T9
RH_9
T_out
Press_mm_hg
RH_out
Windspeed
Visibility
Tdewpoint
rv1
rv2

Data Preprocessing

3.1 Missing Value Analysis

The original dataset contains no missing values or duplicate records. However, creating lag, rolling, and difference features introduces missing values in the initial rows because previous observations are unavailable. These rows were removed before model training.

3.2 Outlier Detection and Treatment

Outliers in appliance energy consumption were handled using the Interquartile Range (IQR) method with a multiplier of 1.5. The outlier boundaries were calculated using training data only. Values outside these boundaries were clipped rather than removing observations, preserving the continuous time series.

3.3 Data Scaling and Normalization

StandardScaler was applied to both input features and the target variable. The scalers were fitted only on the training data and then applied to the validation and test sets to prevent data leakage.

3.4 Data Splitting

The dataset was divided chronologically without shuffling to preserve the time series order.

Training - 70%
Validation - 15%
Testing - 15%

Feature Engineering

4.1 Time-Based Features

Hour of day and day of week were extracted from timestamps to capture daily and weekly consumption patterns. Sine and cosine transformations were also applied to represent their cyclical nature.

4.2 Rolling Window Features

Rolling mean and standard deviation features were created using windows of 3, 6, 18, 36, and 144 observations. These features capture recent energy consumption trends and variations over different time periods.

4.3 Lagged Features

Lag features were created using previous appliance consumption values at 1, 2, 3, 4, 5, 6, 18, 36, and 144 time steps. These allow models to learn from historical energy usage patterns.

4.4 Interaction Features

4.5 Domain-Specific Features (if applicable)

4.6 Feature Selection

Feature selection experiments were considered to identify useful predictors and reduce unnecessary features. The final feature set will be determined based on validation performance.

Model Design

1. Baseline Models:

Linear Regression
Ridge regression

XGBoost
Catboost
Lightboost

2. Deep Learning Model Design:

LSTM

GRU
CNN-LSTM

3. Activation Functions and Optimizers:

Activation Function - GeLU
Optimizer - AdamW

4. Loss Functions:

Mean average error (MAE)
Mean squared error (MSE)
Root mean square error (RMSE)
Huber Loss

5. Model Complexity and Regularization:

Model complexity - Different hidden sizes, layer counts, and sequence lengths were configurable for experimentation.

Regularization - Dropout, Weight decay

6. Model Training and Evaluation

1. Training:

2. Evaluation Metrics:

Mean average error (MAE)
Mean squared error (MSE)
Root mean square error (RMSE)
Huber Loss

3. Performance Assessment:

The performance of baseline and deep learning models was compared using MAE, MSE, RMSE, and R^2.

4. Visualization:

7. Model Optimization

7.1 Hyperparameter Tuning

7.2 Regularization Experiments

7.3 Early Stopping

7.4 Optimization Outcomes

Results and Discussion

8.1 Model Performance Comparison

8.2 Predicted vs. Actual Values

8.3 Residual and Error Analysis

8.4 Discussion of Findings

Challenges and Solutions

Conclusion

References